

1. Introduction to Excel

Microsoft Excel is a spreadsheet application used to store, organize, analyze, and visualize data. It is widely used in business, finance, data analysis, and reporting.

Uses of Excel

- Data Entry
- Data Cleaning
- Calculations
- Reporting
- Data Analysis
- Dashboard Creation

2. Workbook and Worksheet

Workbook

A workbook is an Excel file that contains one or more worksheets.

Worksheet

A worksheet is a single spreadsheet within a workbook where data is stored and analyzed.

3. Rows, Columns, and Cells

Row

Horizontal arrangement of cells identified by numbers.

Example:

1, 2, 3, 4...

Column

Vertical arrangement of cells identified by letters.

Example:

A, B, C, D...

Cell

Intersection of a row and a column.

Examples:

A1
B5
C10

4. Data Entry and Editing

Excel allows users to enter and modify different types of data.

Types of Data

- Text
- Numbers
- Dates
- Formulas

Common Operations

- Copy (Ctrl + C)
- Paste (Ctrl + V)
- Cut (Ctrl + X)
- Undo (Ctrl + Z)
- Redo (Ctrl + Y)

5. Cell Formatting

Formatting improves readability and presentation of data.

Formatting Features

- Font Size
- Font Style
- Bold
- Italic
- Underline
- Borders
- Cell Colors
- Text Alignment

Number Formats

- General
- Currency
- Percentage
- Date
- Time

6. AutoFill and Flash Fill

AutoFill

Used to quickly fill a series of numbers, dates, or patterns.

Example:

1,2,3,4,5...

Flash Fill

Automatically detects patterns and fills data accordingly.

Shortcut:

Ctrl + E

7. Sorting Data

Sorting arranges data in a specific order.

Types of Sorting

- Ascending (A to Z)
- Descending (Z to A)
- Smallest to Largest
- Largest to Smallest

Benefits

- Easy data analysis
- Better organization

8. Filtering Data

Filtering displays only the records that meet selected criteria.

Example

Show:

- Only Hyderabad employees
- Sales greater than ₹5,000
- Female customers

Path:

Data → Filter

9. Find and Replace

Used to locate and modify values quickly.

Shortcuts

Find:

Ctrl + F

Replace:

Ctrl + H

10. Basic Excel Functions

SUM()

Adds values.

Example:

=SUM(A1:A10)

AVERAGE()

Calculates average value.

Example:

=AVERAGE(A1:A10)

COUNT()

Counts numeric cells.

Example:

=COUNT(A1:A10)

MAX()

Returns highest value.

Example:

=MAX(A1:A10)

MIN()

Returns lowest value.

Example:

=MIN(A1:A10)

COUNTIF()

Counts cells based on a condition.

Example:

=COUNTIF(E2:E1095,">5000")

Counts all values greater than 5000.

11. Cell References

Relative Reference

Changes automatically when copied.

Example:

=A1+B1

Absolute Reference

Remains fixed when copied.

Example:

=\$A\$1+\$B\$1

12. Freeze Panes

Freeze Panes keeps selected rows or columns visible while scrolling.

Path:

View → Freeze Panes

Advantages

- Easy navigation
- Better visibility of headers

13. Remove Duplicates

Removes repeated records from a dataset.

Path:

Data → Remove Duplicates

Benefits

- Cleaner data
- Improved accuracy

14. Data Validation

Restricts user input to specific values.

Example

Allow only:

- Yes / No
- Male / Female
- Specific date ranges

Path:

Data → Data Validation